CNC PROGRAM SHEET

Part: RKSE-004-057 | Rev: A | Date: 2026-05-15

Programmer: CNC-AI

Material: ALUMINIUM

Max RPM: 10000 Coolant: through spindle

Setups: 24

Part envelope: 18.4 x 40.1 x 62.3 mm

COORDINATE SYSTEM & WORK ZERO (G54)

Work zero convention	--	**Part height (Y)**	-- mm
CAD up-axis	--	**X range (machine)**	-- to -- mm
Work zero in CAD	X=0 Y=0 Z=0	**Z range (machine)**	-- to -- mm
G54 meaning	Set controller G54 offset so that X0 Y0 Z0 = top face center of part	**Safe rapid Y**	10 mm above work zero

TOOL LIST

T#	Tool ID	Description	Dia (mm)	Manufacturer	Grade	Used for
T01	CD_60_2MM	60-degree combined center drill, 2mm tip, 5mm body	2.0	Sandvik Coromant	H10F	spot_drill
T02	DRILL_2.0MM	Solid carbide jobber drill 2.0mm	2.0	Sandvik Coromant	H10F	twist_drill
T03	SPOT_090_4MM	90-degree solid carbide spot drill, 4mm shank	4.0	Sandvik Coromant	H10F	spot_drill
T04	DRILL_6.0MM	Solid carbide jobber drill 6.0mm	6.0	Sandvik Coromant	H10F	twist_drill
T05	CHAM_90_6MM	90-degree solid carbide chamfer mill 6mm, 2-flute	6.0	Sandvik Coromant	1630	chamfer_mill
T06	FACEMILL_40MM	40mm indexable face mill â€” 6 inserts	40.0	Sandvik Coromant		face_mill

SETUP 1 -- +Y approach (top — default VMC position)

PRINCIPAL | 8 clusters | 11 operations

Fixture	Standard vise or fixture plate, jaws on side faces.
Description	Standard top-face setup. Clamp part with top face accessible to spindle. Machine all features in this setup before repositioning.
Rotation	None -- standard orientation
WCS	G54
Origin X/Y/Z	X=CENTER (X=-182.941mm, 9.19mm from xmin edge) Y=CENTER (Z=0.000mm, 31.15mm from zmin edge) Z=TOP
Face (W x H)	18.4 x 62.3 mm
Max depth	40.1 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T01	2.0	0	1.0	10000	300	1.0	--	--	00:01
2	ALU 2 DRILL	T02	2.0	0	8.62	10000	320	8.625	--	--	00:04
3	ALU CENTER DRILL	T01	2.0	0	1.0	10000	300	1.0	--	--	00:01
4	ALU 2 DRILL	T02	2.0	0	8.62	10000	320	8.625	--	--	00:04
5	ALU COUNTERSINK COUNTERSINK	T??	--	0	--	--	--	--	--	--	00:00
6	ALU COUNTERSINK COUNTERSINK	T??	--	0	--	--	--	--	--	--	00:00
7	ALU COUNTERSINK COUNTERSINK	T??	--	0	--	--	--	--	--	--	00:00
8	ALU COUNTERSINK COUNTERSINK	T??	--	0	--	--	--	--	--	--	00:00
9	ALU COUNTERSINK COUNTERSINK	T??	--	0	--	--	--	--	--	--	00:00
10	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
11	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00

ESTIMATED SETUP TIME: 00:26 (includes tool changes)

SETUP 2 -- -Y approach (bottom — part flipped upside down)

PRINCIPAL | 5 clusters | 8 operations

Fixture	Step-jaw vise or fixture plate with bosses as datums.
Description	Flipped setup — part inverted so bottom face is accessible to spindle. Clamp on previously machined top features or use step-jaw vise.
Rotation	None -- standard orientation
WCS	G55
Origin X/Y/Z	X=CENTER (X=-182.941mm, 9.19mm from xmin edge) Y=CENTER (Z=0.000mm, 31.15mm from zmin edge) Z=TOP
Face (W x H)	18.4 x 62.3 mm
Max depth	40.1 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T03	4.0	0	2.0	10000	600	2.0	--	--	00:01
2	ALU 6 DRILL	T04	6.0	0	5.0	9550	716	5.0	--	--	00:01
3	ALU CENTER DRILL	T03	4.0	0	2.0	10000	600	2.0	--	--	00:01
4	ALU 6 DRILL	T04	6.0	0	2.0	9550	716	2.0	--	--	00:01
5	ALU CENTER DRILL	T03	4.0	0	2.0	10000	600	2.0	--	--	00:01
6	ALU 6 DRILL	T04	6.0	0	5.0	9550	716	5.0	--	--	00:01
7	ALU COUNTERSINK COUNTERSINK	T??	--	0	--	--	--	--	--	--	00:00
8	ALU COUNTERSINK COUNTERSINK	T??	--	0	--	--	--	--	--	--	00:00

ESTIMATED SETUP TIME: 00:20 (includes tool changes)

SETUP 3 -- +X approach (right side face)

PRINCIPAL | 5 clusters | 7 operations

Fixture	90° angle plate or tombstone fixture.
Description	Right-side face setup. Rotate part 90° so right face is accessible. Clamp on bottom and left faces.
Rotation	None -- standard orientation
WCS	G56
Origin X/Y/Z	X=CENTER (Y=-42.586mm) Y=CENTER (Z=0.000mm) Z=TOP
Face (W x H)	40.1 x 62.3 mm
Max depth	18.4 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T01	2.0	0	1.0	10000	300	1.0	--	--	00:01
2	ALU 2 DRILL	T02	2.0	0	3.87	10000	320	3.866	--	--	00:01
3	ALU CENTER DRILL	T03	4.0	0	2.0	10000	600	2.0	--	--	00:01
4	ALU 6 DRILL	T04	6.0	0	4.0	9550	716	4.0	--	--	00:01
5	ALU COUNTERSINK COUNTERSINK	T??	--	0	--	--	--	--	--	--	00:00
6	ALU COUNTERSINK COUNTERSINK	T??	--	0	--	--	--	--	--	--	00:00
7	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04

ESTIMATED SETUP TIME: 00:47 (includes tool changes)

SETUP 4 -- -X approach (left side face)

PRINCIPAL | 1 clusters | 2 operations

Fixture	90° angle plate or tombstone fixture.
Description	Left-side face setup. Rotate part 90° so left face is accessible. Clamp on bottom and right faces.
Rotation	None -- standard orientation
WCS	G57
Origin X/Y/Z	X=CENTER (Y=-42.586mm) Y=CENTER (Z=0.000mm) Z=TOP
Face (W x H)	40.1 x 62.3 mm
Max depth	18.4 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T01	2.0	0	1.0	10000	300	1.0	--	--	00:01
2	ALU 2 DRILL	T02	2.0	0	3.87	10000	320	3.866	--	--	00:01

ESTIMATED SETUP TIME: 00:17 (includes tool changes)

SETUP 5 -- +Z approach (front face)

PRINCIPAL | 1 clusters | 2 operations

Fixture	90° angle plate or tombstone fixture.
Description	Front-face setup. Rotate part 90° so front face is accessible to spindle.
Rotation	None -- standard orientation
WCS	G58
Origin X/Y/Z	X=CENTER (X=-182.941mm) Y=CENTER (Y=-42.586mm) Z=TOP
Face (W x H)	18.4 x 40.1 mm
Max depth	62.3 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T03	4.0	0	2.0	10000	600	2.0	--	--	00:01
2	ALU 6 DRILL	T04	6.0	0	4.0	9550	716	4.0	--	--	00:01

ESTIMATED SETUP TIME: 00:17 (includes tool changes)

SETUP 6 -- -Z approach (rear face)

PRINCIPAL | 1 clusters | 1 operations

Fixture	90° angle plate or tombstone fixture.
Description	Rear-face setup. Rotate part 90° so rear face is accessible to spindle.
Rotation	None -- standard orientation
WCS	G59
Origin X/Y/Z	X=CENTER (X=-182.941mm) Y=CENTER (Y=-42.586mm) Z=TOP
Face (W x H)	18.4 x 40.1 mm
Max depth	62.3 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 40 FACEMILL FACE FINISH	T06	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04

ESTIMATED SETUP TIME: 00:12 (includes tool changes)

SETUP 7 -- angled approach [-0.941, -0.0, -0.339]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 90.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 90.0° around -Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	90.0 deg around -Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER (Y=-42.586mm) Y=CENTER (Z=0.000mm) Z=TOP
Face (W x H)	40.1 x 62.3 mm
Max depth	18.4 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 8 -- angled approach [0.9, -0.0, -0.436]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 90.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 90.0° around +Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	90.0 deg around +Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER (Y=-42.586mm) Y=CENTER (Z=0.000mm) Z=TOP
Face (W x H)	40.1 x 62.3 mm
Max depth	18.4 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 9 -- angled approach [0.9, -0.0, 0.436]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 90.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 90.0° around +Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	90.0 deg around +Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER (Y=-42.586mm) Y=CENTER (Z=0.000mm) Z=TOP
Face (W x H)	40.1 x 62.3 mm
Max depth	18.4 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 10 -- angled approach [-0.941, -0.0, 0.339]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 90.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 90.0° around -Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	90.0 deg around -Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER (Y=-42.586mm) Y=CENTER (Z=0.000mm) Z=TOP
Face (W x H)	40.1 x 62.3 mm
Max depth	18.4 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 11 -- angled approach [0.637, 0.707, -0.308]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around +Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around +Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 12 -- angled approach [0.707, 0.707, -0.0]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around +Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around +Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 13 -- angled approach [0.637, 0.707, 0.308]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around +Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around +Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 14 -- angled approach [-0.665, 0.707, -0.24]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around -Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around -Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 15 -- angled approach [-0.707, 0.707, -0.0]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around -Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around -Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 16 -- angled approach [-0.665, 0.707, 0.24]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around -Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around -Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 17 -- angled approach [-0.637, 0.707, -0.308]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around -Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around -Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 18 -- angled approach [0.665, 0.707, -0.24]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around +Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around +Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 19 -- angled approach [0.707, -0.0, 0.707]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 90.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 90.0° around compound axis [-0.707, 0.0, 0.707] — use sine plate or 5-axis fixture from the default orientation. Ve
Rotation required	90.0 deg around compound axis [-0.707, 0.0, 0.707] — use sine plate or 5-axis fixture -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 20 -- angled approach [-0.0, -0.707, -0.707]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 135.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 135.0° around +X axis (A-axis tilt) from the default orientation. Verify with dial indicator before machining. All featu
Rotation required	135.0 deg around +X axis (A-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 21 -- angled approach [0.707, -0.0, -0.707]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 90.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 90.0° around compound axis [0.707, 0.0, 0.707] — use sine plate or 5-axis fixture from the default orientation. Ver
Rotation required	90.0 deg around compound axis [0.707, 0.0, 0.707] — use sine plate or 5-axis fixture -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 22 -- angled approach [-0.0, -0.707, 0.707]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 135.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 135.0° around -X axis (A-axis tilt) from the default orientation. Verify with dial indicator before machining. All featu
Rotation required	135.0 deg around -X axis (A-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 23 -- angled approach [0.665, 0.707, 0.24]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around +Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around +Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

SETUP 24 -- angled approach [-0.637, 0.707, 0.308]

ANGLED | 1 clusters | 1 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around -Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around -Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 6.0 CHAMFER MILL CHAMFER	T05	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 00:18 (includes tool changes)

WARNINGS & FLAGGED ITEMS

Type	Cluster	Operation	Detail
RPM CAPPED	C0	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C0	twist_drill	SUBSTITUTION: 6.4mm is non-standard. Using nearest standard size 6.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C1	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C1	twist_drill	SUBSTITUTION: 6.4mm is non-standard. Using nearest standard size 6.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C2	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C2	twist_drill	SUBSTITUTION: 6.4mm is non-standard. Using nearest standard size 6.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C3	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C3	twist_drill	Effective Vc = 62.8 m/min (reduced from 120 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C4	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C4	twist_drill	Effective Vc = 62.8 m/min (reduced from 120 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C5	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C5	twist_drill	Effective Vc = 62.8 m/min (reduced from 120 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C6	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C6	twist_drill	Effective Vc = 62.8 m/min (reduced from 120 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C7	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C7	twist_drill	SUBSTITUTION: 6.4mm is non-standard. Using nearest standard size 6.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C8	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C8	twist_drill	SUBSTITUTION: 6.4mm is non-standard. Using nearest standard size 6.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C21	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C22	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C24	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C25	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C27	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C28	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C29	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C30	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C31	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C32	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C33	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C34	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C35	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C36	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C37	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C38	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C39	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C40	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.